

M01 - Introduction

...

The Development Environment

The Development Environment

1. Visual Studio Code (VSC)
2. Git & GitHub

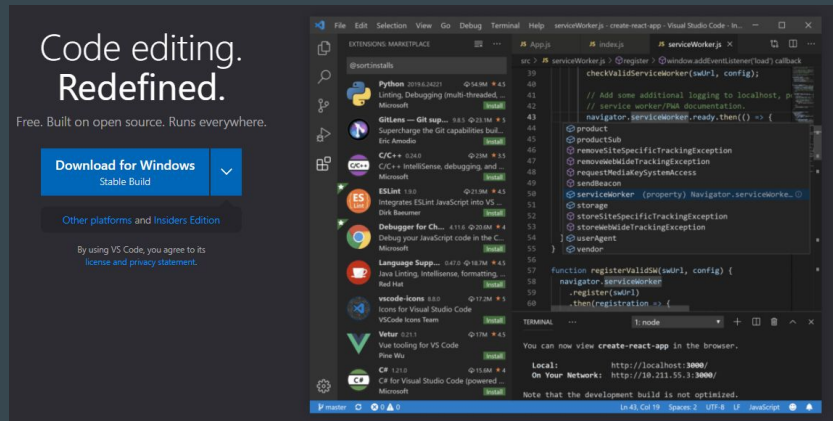
The Development Environment

1. Visual Studio Code (VSC)

The Development Environment

1. Visual Studio Code

- Code Editor from Microsoft
- Link: <https://code.visualstudio.com/>
- Docs: <https://code.visualstudio.com/docs>



The Development Environment

1. Visual Studio Code

EXPLORER
(CTRL+B)

SEARCH

SOURCE
CONTROL

DEBUG

EXTENSIONS

MENU BAR

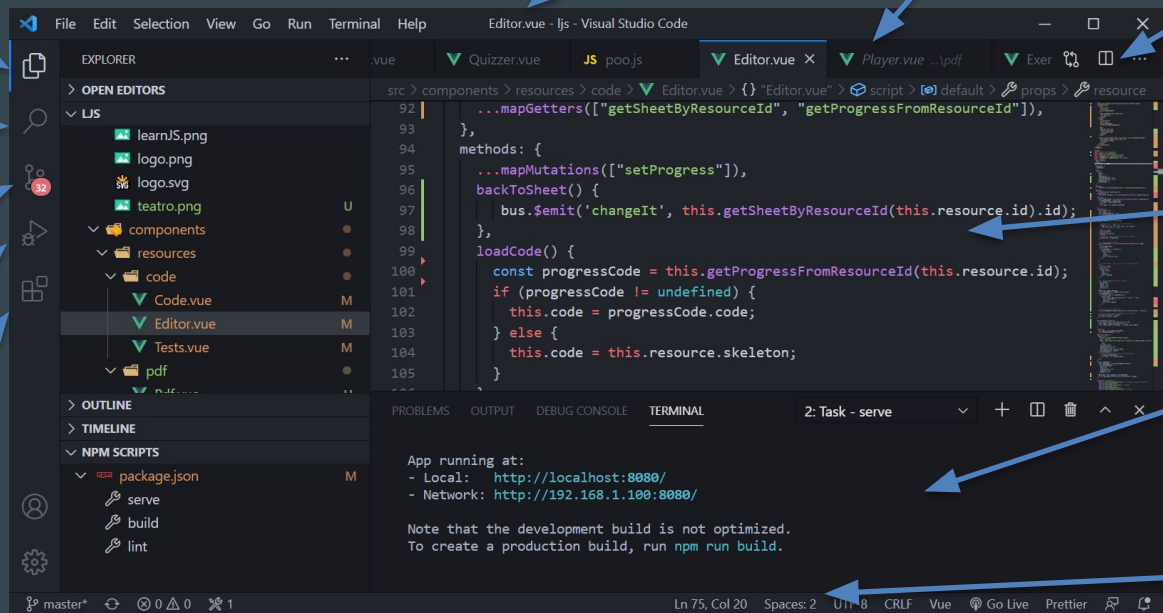
FILE TABS

SPLIT EDITOR

CODE AREA

TERMINAL
(CTRL+`)

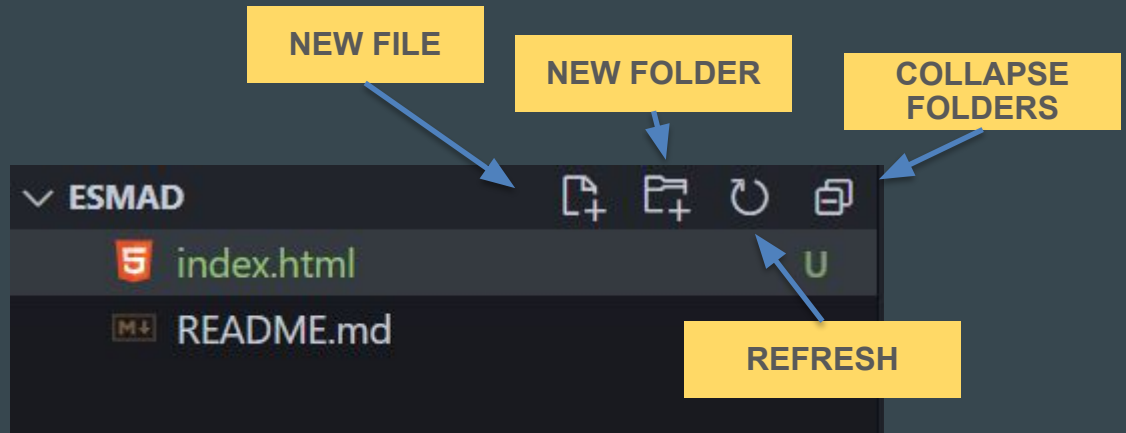
STATUS BAR



The Development Environment

1. Visual Studio Code

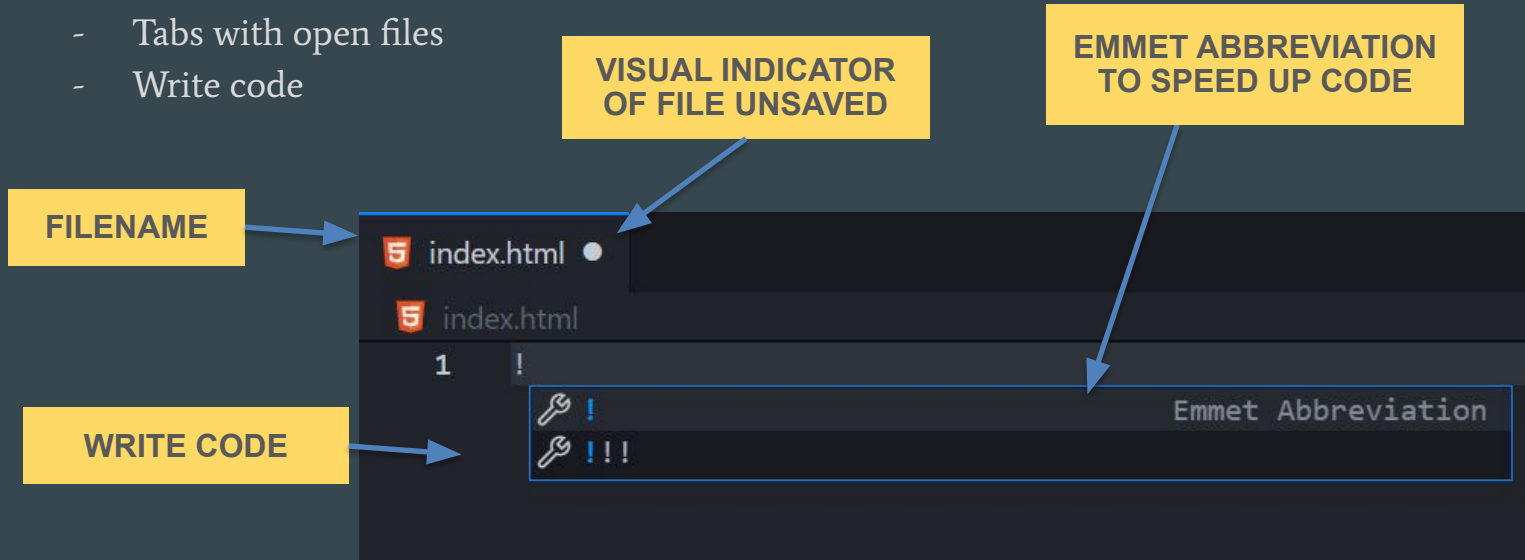
- File Explorer (Ctrl+Shift+E)



The Development Environment

1. Visual Studio Code

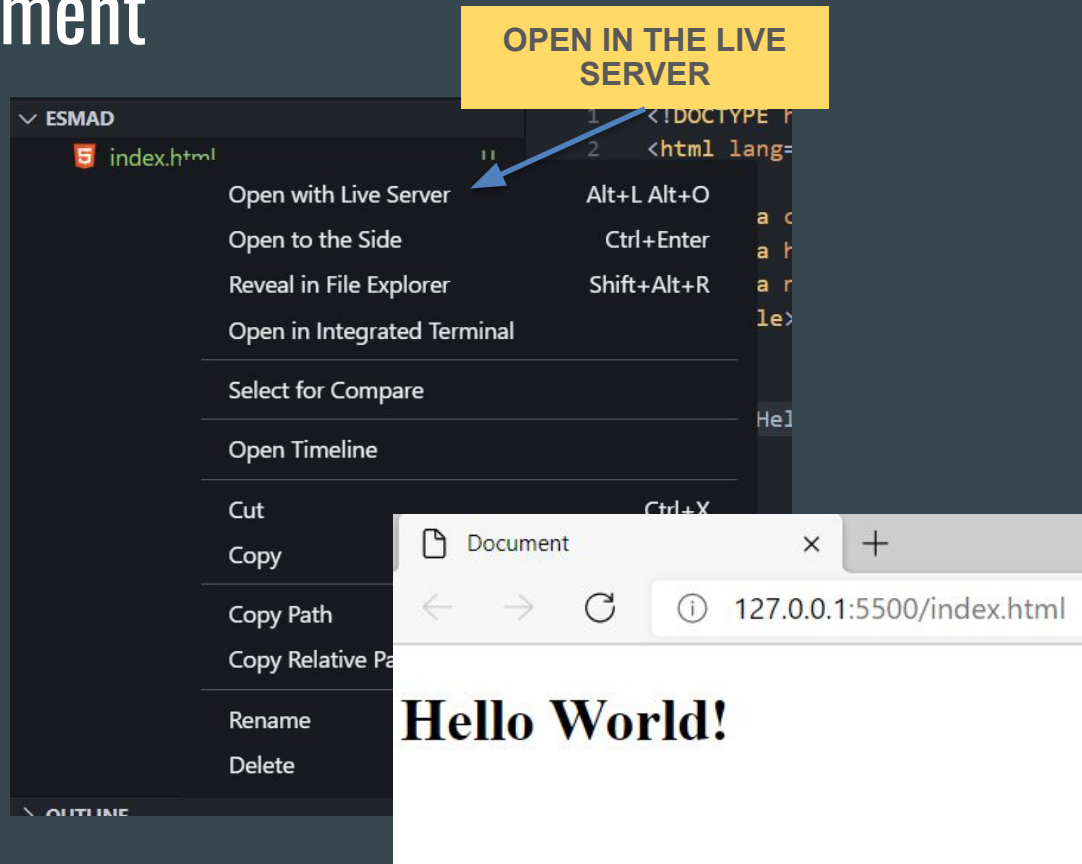
- Code Editor
 - Tabs with open files
 - Write code



The Development Environment

1. Visual Studio Code

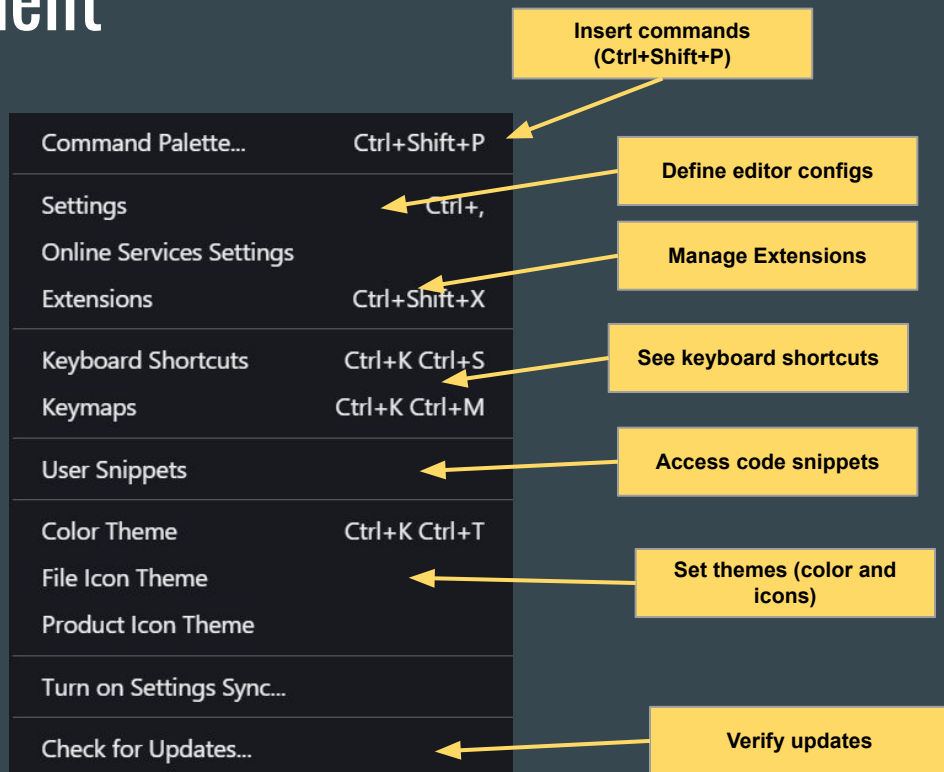
- Code Editor
 - Right mouse button
 - to open file in several ways
 - to copy/cut/rename/delete



The Development Environment

1. Visual Studio Code

- Preferences:
 - File->Preferences
 - Gear icon (lower left corner)



The Development Environment

1. Visual Studio Code

- Themes
 - Global color patterns in the editor
 - Choose a theme
 - **File->Preferences->Color Theme** (Ctrl+K+T)
 - Most popular themes:
 - Material Theme
 - Dracula
 - Monokai
 - Firewatch
- You can install new themes with: **Install Additional Color Themes**

Select Color Theme (Up/Down Keys to Preview)

Solarized Light

Abyss

Dark (Visual Studio)

Dark+ (default dark)

Kimbie Dark

Monokai

Monokai Dimmed

Night Owl

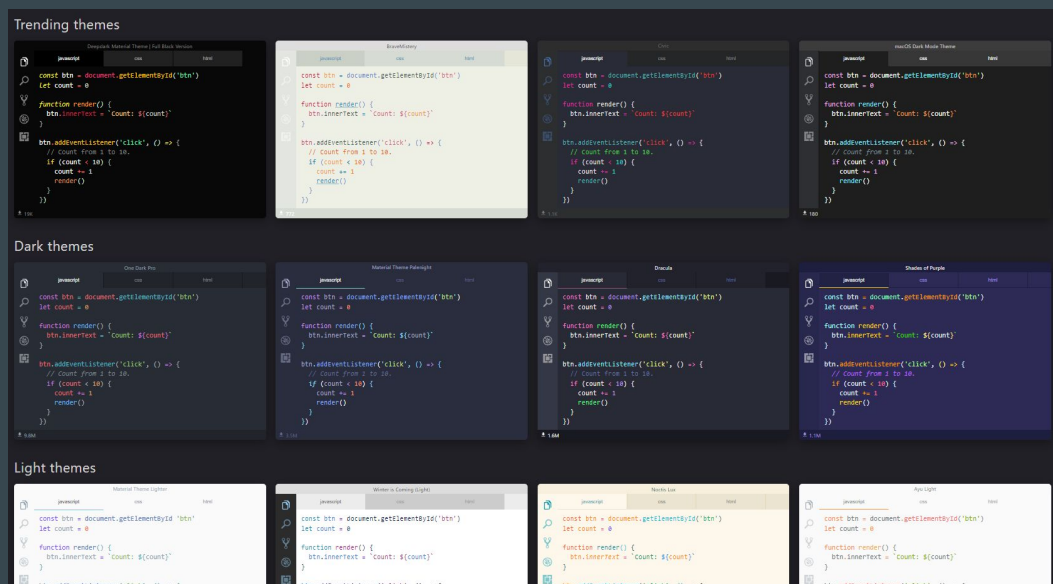
Night Owl (No Italics)

Plastic

The Development Environment

1. Visual Studio Code (themes)

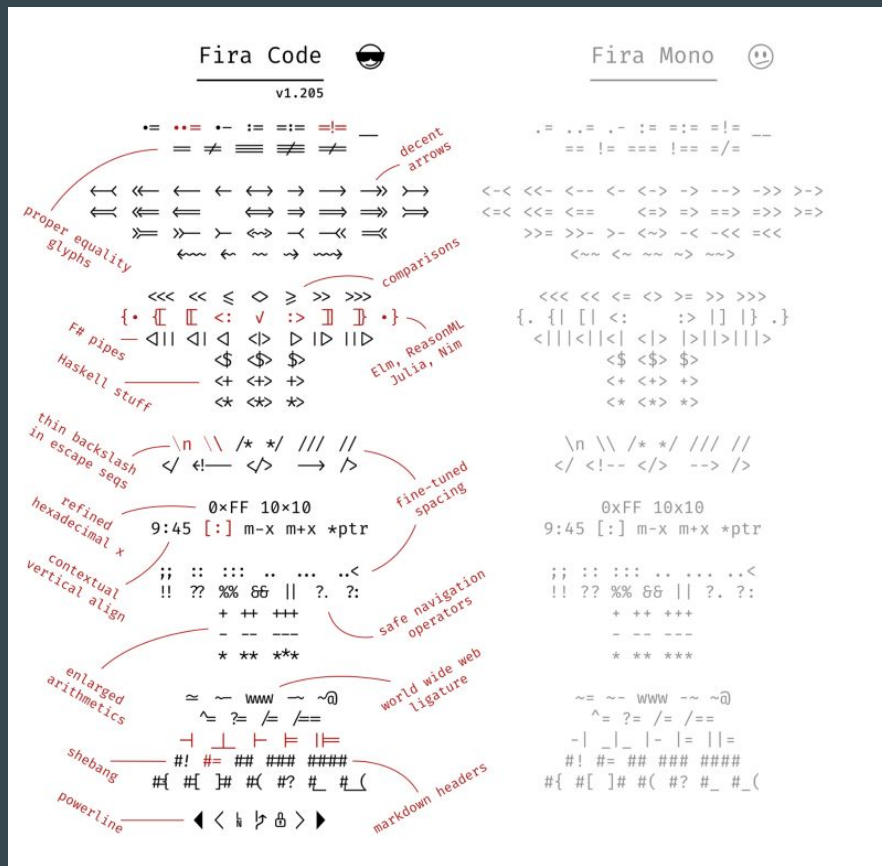
- Visit the URL <https://vscodethemes.com> to see most popular themes!



The Development Environment

1. Visual Studio Code

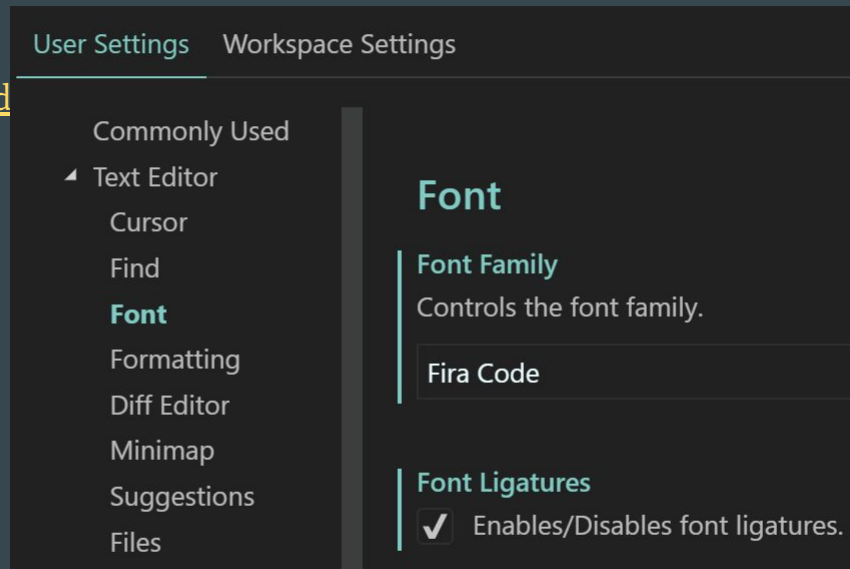
- Fonts
 - Use of **Fira Code**
 - Free monospaced font with programming ligatures



The Development Environment

1. Visual Studio Code

- Fonts
 - Use of **Fira Code**
 - Install: <https://github.com/tonsky/FiraCode>
 - Activation through Settings (**Ctrl+,**)



The Development Environment

1. Visual Studio Code

- Fonts
 - Use of Fira Code

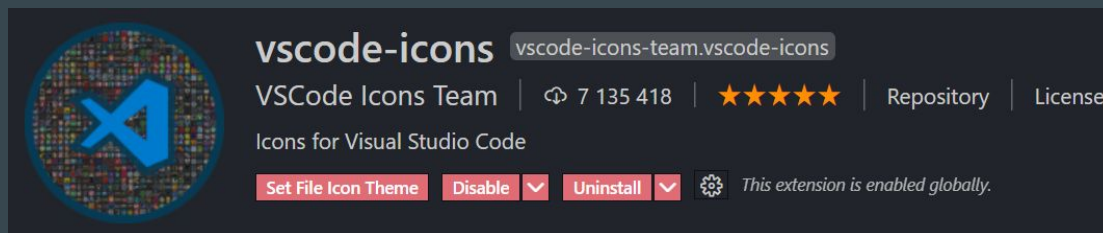
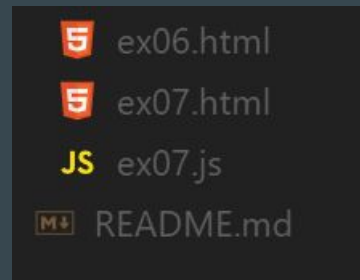
```
function isMovie(newTitle) {  
  for (const movie of movies) {  
    if (movie.title === newTitle) {  
      return true;  
    }  
  }  
  return false;  
}
```

```
function isMovie(newTitle) {  
  for (const movie of movies) {  
    if (movie.title === newTitle) {  
      return true;  
    }  
  }  
  return false;  
}
```

The Development Environment

1. Visual Studio Code

- Icons
 - The presence of multiple files can make the explorer difficult to read and manage
 - Use of typified icons
 - Activation: **File > Preferences > File Icon Theme > VSCode Icons**



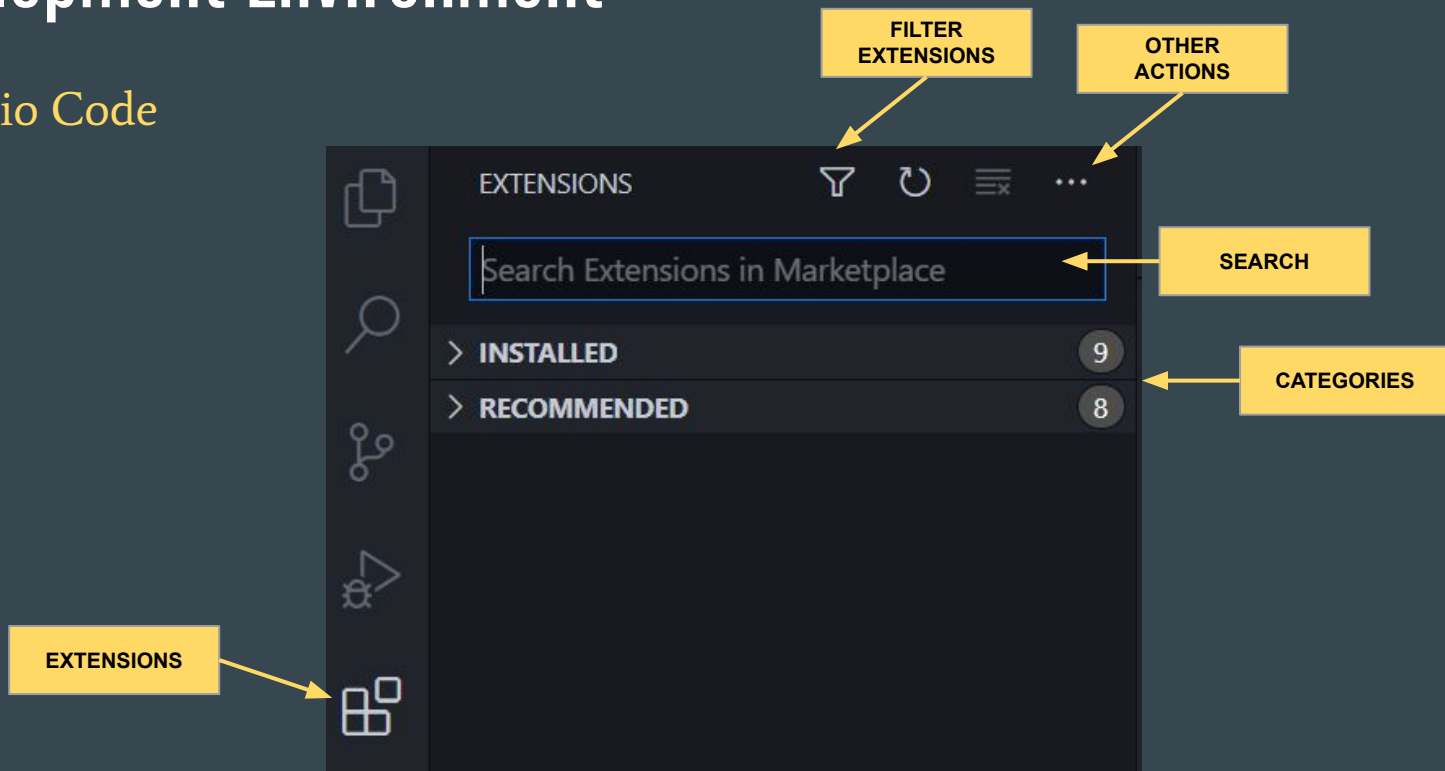
The Development Environment

1. Visual Studio Code

- Extensions
 - Utilities that you can install in VSC
- Main goals:
 - increment productivity
 - make the code more readable
 - speed up code edition
 - foster cooperation

The Development Environment

1. Visual Studio Code



The Development Environment

1. Visual Studio Code

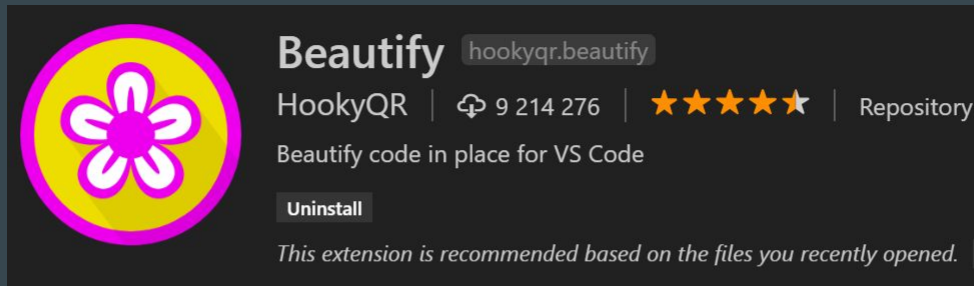
- **Formatting:** Beautify, ...
- **Manage:** Bookmarks, Todo Highlight, Guides, Image Preview, Color Highlight, ...
- **Productivity:** Live Server, Bootstrap Snippets, Quokka, 30 seconds of code, ...
- **Debugging:** Debugger for Chrome, ESLint, ...
- **Others:** Polacode, ...

The Development Environment

1. Visual Studio Code (extensions)

Beautify:

- By default
 - Manual (Shift+Alt+F)
 - Automatically when saving the file
- Formatting with Beautify
 - More sophisticated
 - Press **F1 Beautify** to beautify a selection or the whole file
- Other options: Prettier

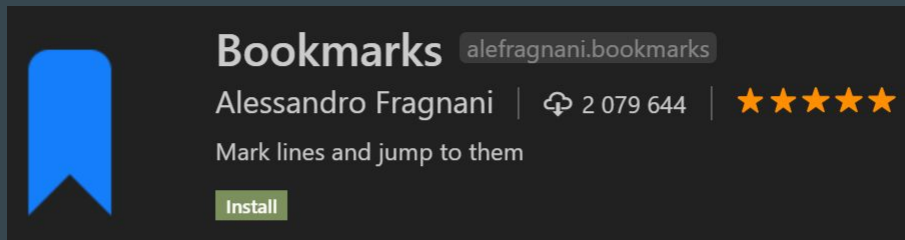


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1. Visual Studio Code (extensions)

Bookmarks:

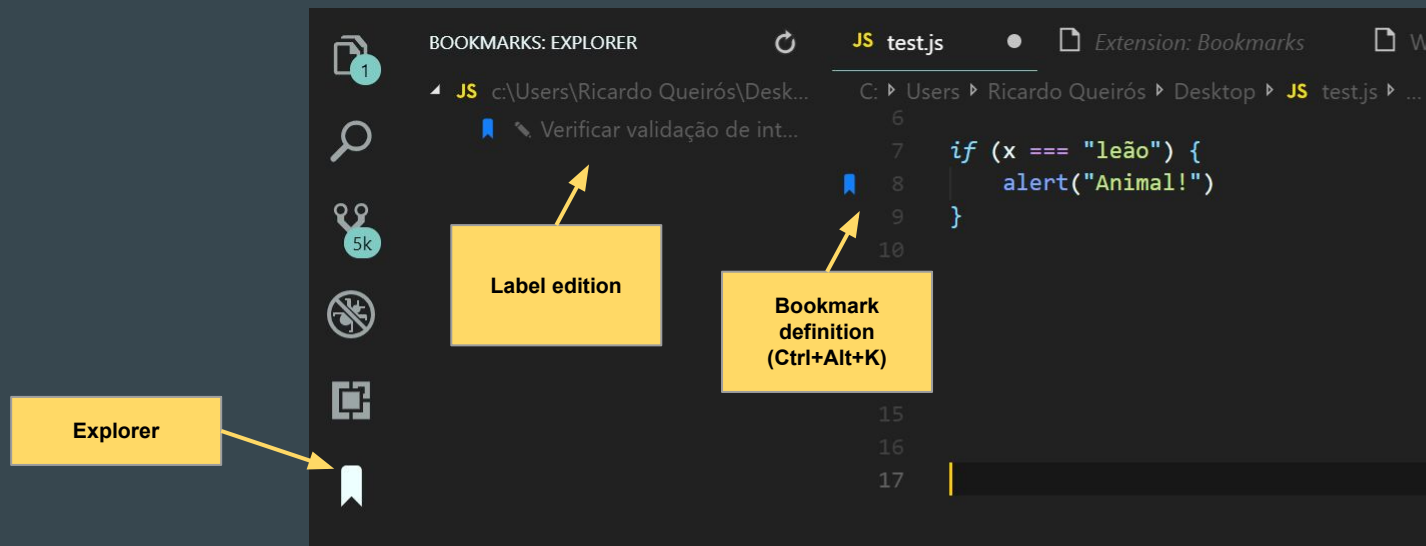
- Style “todo list”
- Definition of marks/tasks



The Development Environment

1. Visual Studio Code (extensions)

Bookmarks:

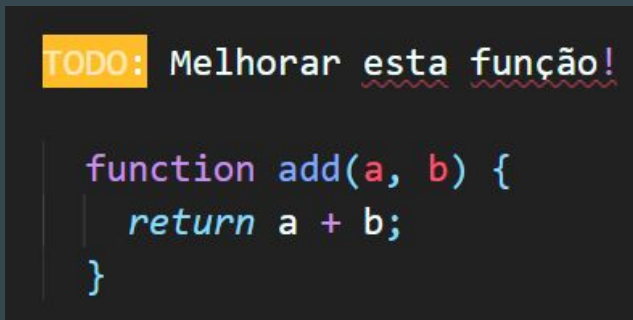
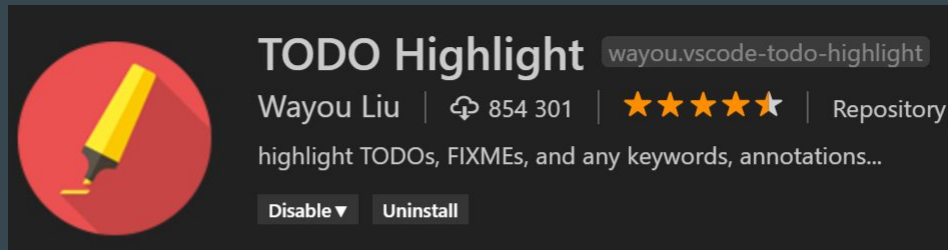


The Development Environment

1. Visual Studio Code (extensions)

Todo Highlight:

- Add TODO marks in the code
- Marks available:
 - TODO
 - FIXME
 - ALL



The Development Environment

1. Visual Studio Code (extensions)

Guides:

- Put guide lines in the code editor
- Great help when we have several nested blocks

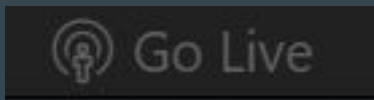
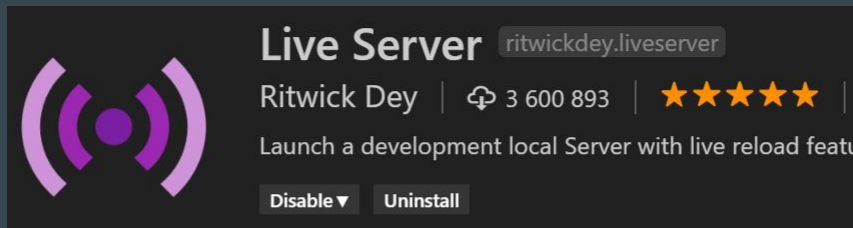


The Development Environment

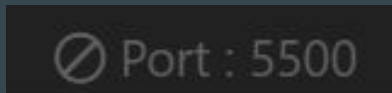
1. Visual Studio Code (extensions)

Live Server:

- Hot-reload
- Server creation which automatically reflects changes in the code
- Server start (local URL: <http://127.0.0.1:5500/>)



- Server end



The Development Environment

1. Visual Studio Code (extensions)

Bootstrap 4, Font awesome 4:

- Code-Snippets for:
 - Bootstrap 4
 - Font awesome 4
 - Font Awesome 5 Free & Pro
- To use bootstrap snippets, open a HTML file and start writing **b4-**, and then press **Ctrl + Space**. A snippets list will appear

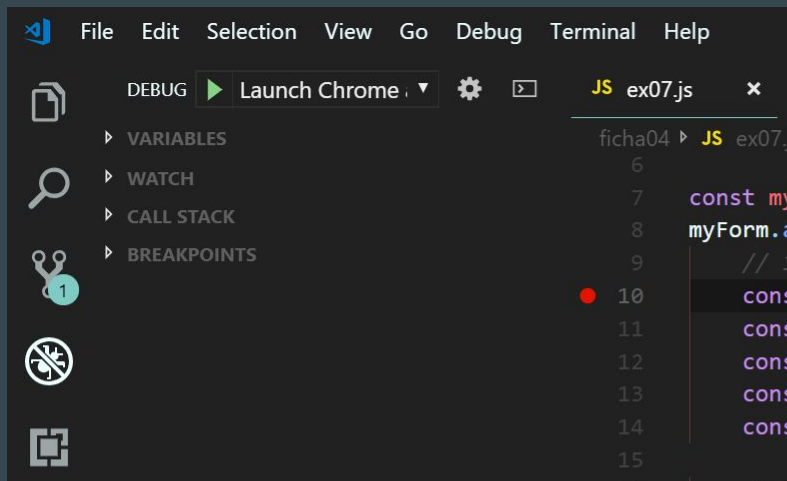
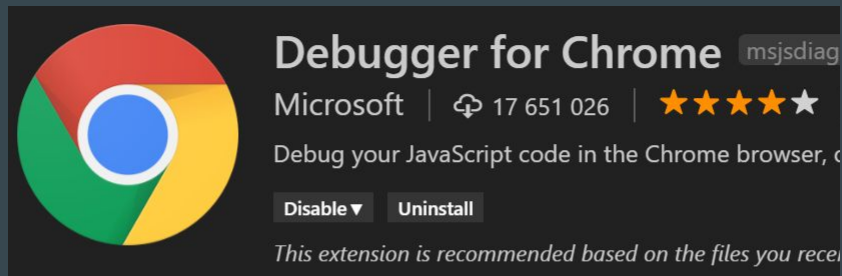


The Development Environment

1. Visual Studio Code (extensions)

Debugger for Chrome:

- Debugging tool integrated in Chrome

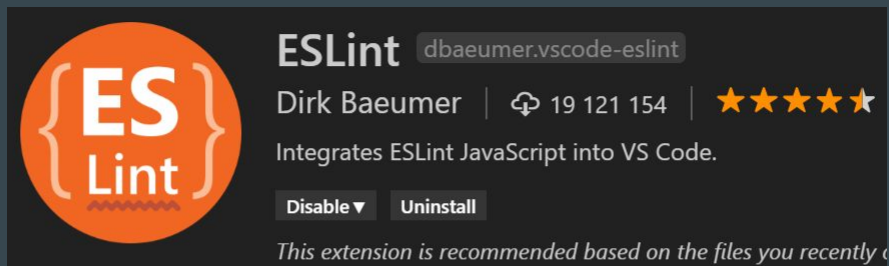


The Development Environment

1. Visual Studio Code (extensions)

ESLint:

- Linter tool
- Analyses and detects issues in code



The Development Environment

1. Visual Studio Code (extensions)

Quokka:

- Playground for JavaScript
- Executes the code while writing and shows several results in the editor

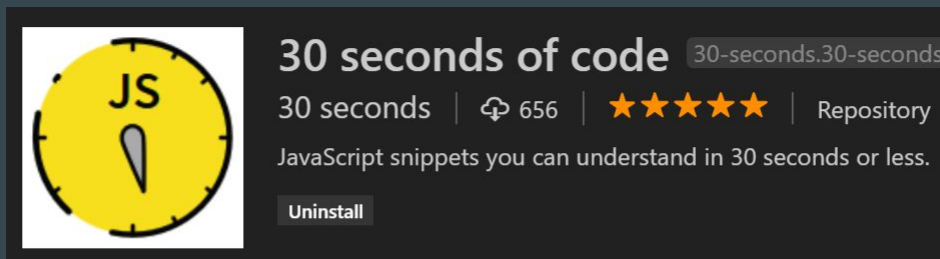


The Development Environment

1. Visual Studio Code (extensions)

30 seconds of code:

- Advanced Code snippets
- Hundred of predefined code snippets
- Invocation with the prefix **30s_**

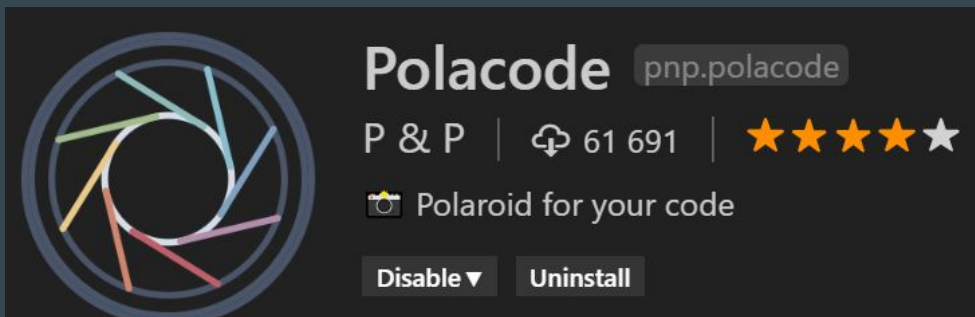


The Development Environment

1. Visual Studio Code (extensions)

Polacode:

- Allows take screenshots from code

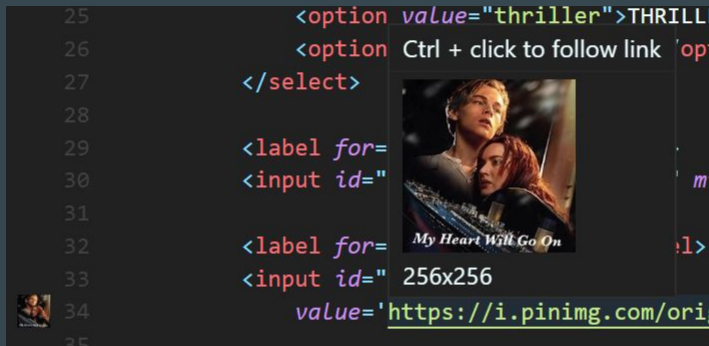
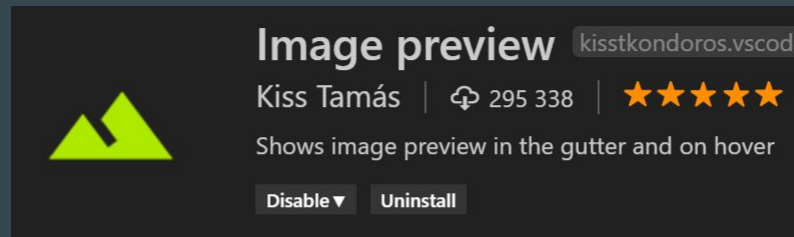


The Development Environment

1. Visual Studio Code (extensions)

Image Preview:

- Show images referenced in the code
 - in the editor gutter
 - when hovering the mouse over the link

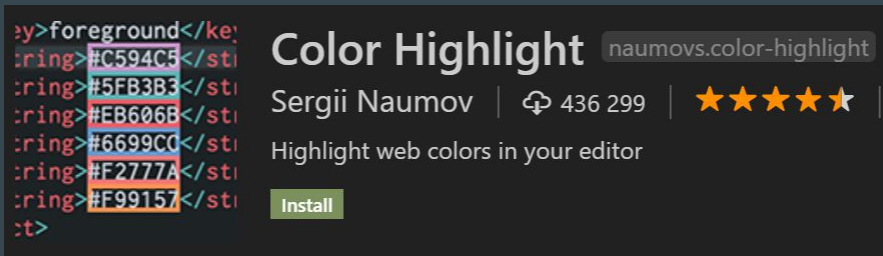


The Development Environment

1. Visual Studio Code (extensions)

Color Highlight:

- Show colors defined in the editor

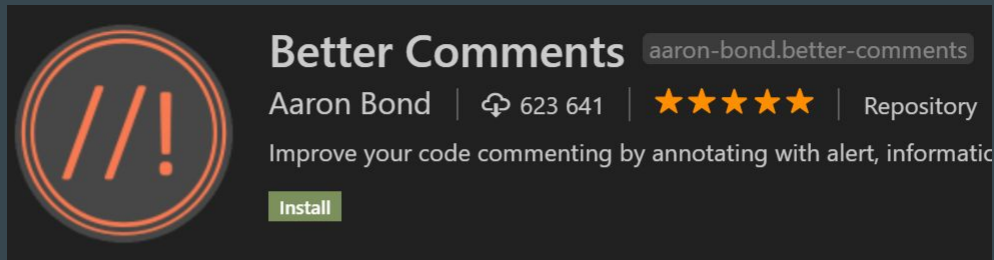


The Development Environment

1. Visual Studio Code (extensions)

Better comments

- Comments more readable
- Categories:
 - Alerts
 - Queries
 - TODOs
 - Highlights



```
///! Isto tem um bug  
///? Será que preciso disto?  
///* Código responsável pelo pedido ao servidor  
///TODO tendo que acabar esta função
```

The Development Environment

2. Git & GitHub

The Development Environment

2. Git & GitHub

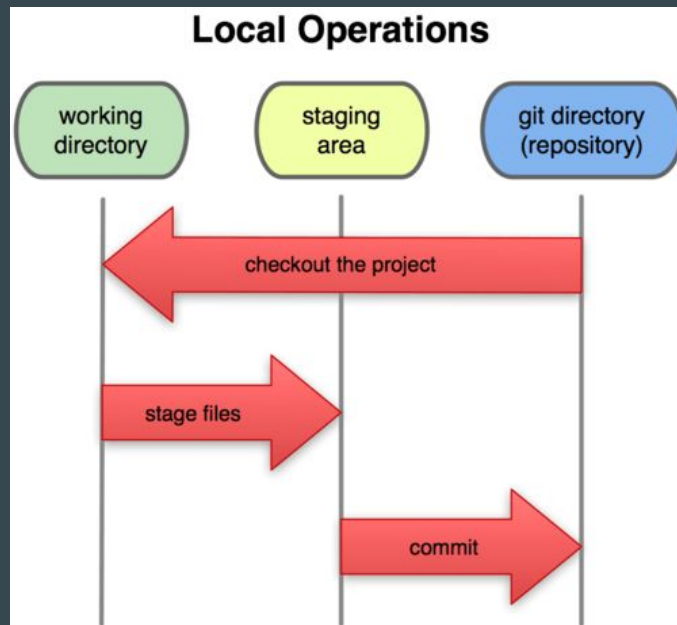
- Git
 - Free and open source distributed version control system
 - Link to install: <https://git-scm.com/>



The Development Environment

2. Git & GitHub

- Git Basic Workflow
 1. Modify files in your **working directory**
 2. Select files adding snapshots from them to you **preparation/staging area**
 3. Make a **commit**, that takes files as they are in their preparation area and stores them permanently in your **git directory**



The Development Environment

2. Git & GitHub

- GitHub
 - Provider of Internet hosting for software development and version control using Git
 - Current owner: Microsoft
 - Main goals:
 - Backup code
 - Manage code versions
 - Foster cooperation among teams
 - Official page: <https://github.com/>



The Development Environment

2. Git & GitHub

Challenge: create a VSC project and integrate it with GitHub

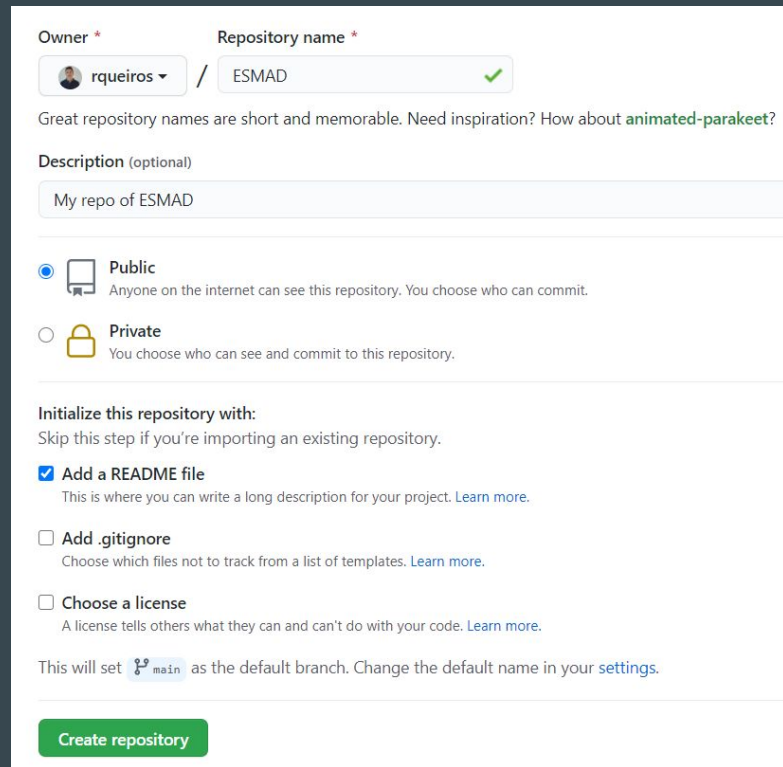
1. Install GIT (<https://git-scm.com>)
2. Create an account at GitHub (<https://github.com>)
3. Read the tutorial (opcional): <https://guides.github.com/activities/hello-world>)
4. Create a repository in GitHub

The Development Environment

2. Git & GitHub

4. Create a repository in GitHub

- Click in “New repository”
- Give a name (for instance: “ESMAD”)
- Give a brief description
- Select the visibility type (select Public)
- Select “Add a README file”
- Click in “Create repository”



The screenshot shows the GitHub 'Create new repository' form. At the top, the 'Owner' is set to 'rqueiros' and the 'Repository name' is 'ESMAD', which is marked as valid with a green checkmark. Below this, a hint suggests repository names should be short and memorable, with 'animated-parakeet?' as an example. The 'Description' field contains 'My repo of ESMAD'. Under the 'Visibility' section, 'Public' is selected, indicating that anyone on the internet can see and commit to the repository. 'Private' is also an option, where only those chosen by the owner can see and commit. The 'Initialize this repository with:' section has three options: 'Add a README file' (checked), 'Add .gitignore', and 'Choose a license'. Each option has a brief explanation and a 'Learn more' link. At the bottom, it states that the default branch will be 'main' and provides a link to 'settings' to change this. A green 'Create repository' button is at the very bottom.

Owner * Repository name *

rqueiros / ESMAD ✓

Great repository names are short and memorable. Need inspiration? How about [animated-parakeet?](#)

Description (optional)

My repo of ESMAD

☒  **Public**
Anyone on the internet can see this repository. You choose who can commit.

☐  **Private**
You choose who can see and commit to this repository.

Initialize this repository with:

Skip this step if you're importing an existing repository.

☒ **Add a README file**
This is where you can write a long description for your project. [Learn more.](#)

☐ **Add .gitignore**
Choose which files not to track from a list of templates. [Learn more.](#)

☐ **Choose a license**
A license tells others what they can and can't do with your code. [Learn more.](#)

This will set  **main** as the default branch. Change the default name in your [settings](#).

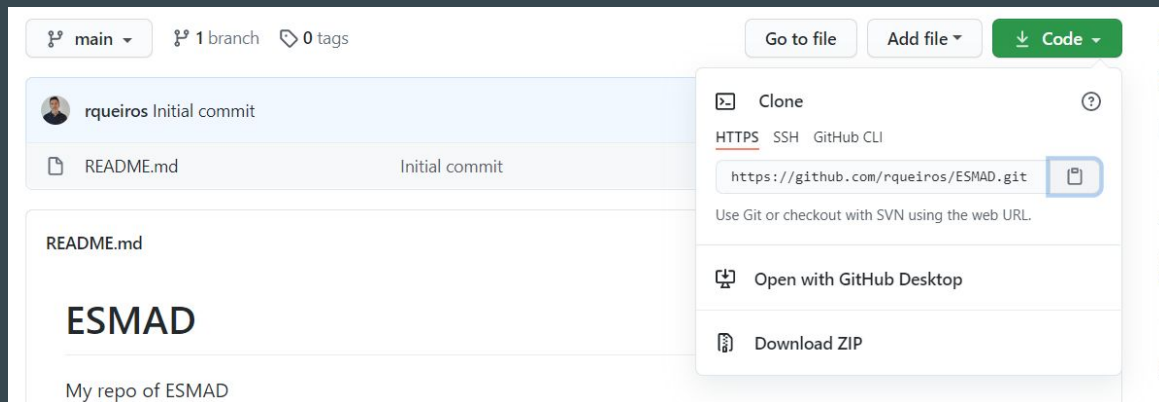
Create repository

The Development Environment

2. Git & GitHub

5. Clone the repo to your computer

- Click in “Code”
- Copy the URL of the repo

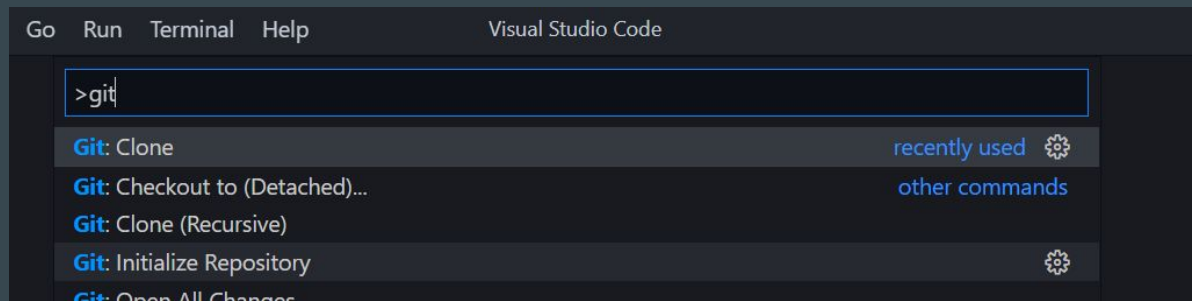


The Development Environment

2. Git & GitHub

6. Integrate the Repo in VSC

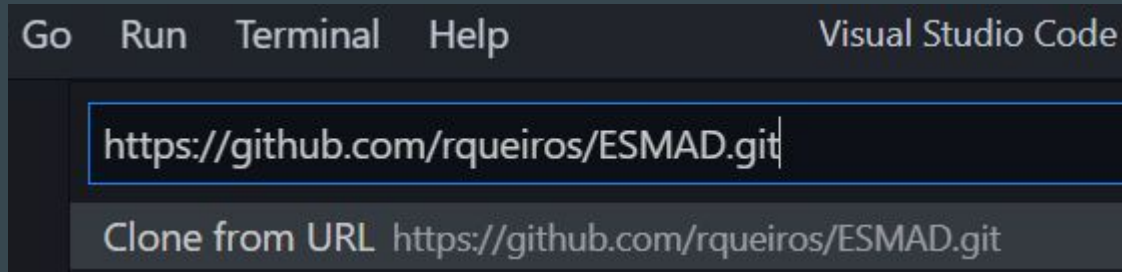
- Open VSC
- Open the command editor (CTRL+SHIFT+P)
- Select Git: Clone



The Development Environment

2. Git & GitHub

6. Integrate the Repo in VSC
 - Put the repo URL and press ENTER

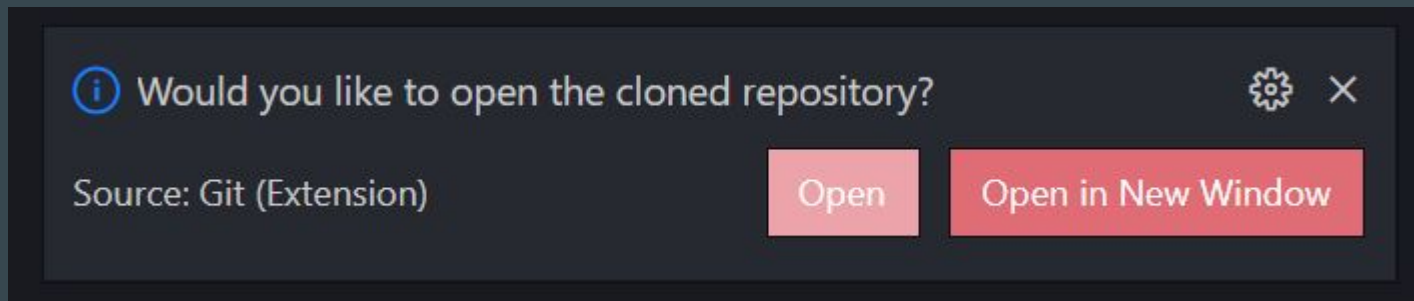


The Development Environment

2. Git & GitHub

6. Integrate the Repo in VSC

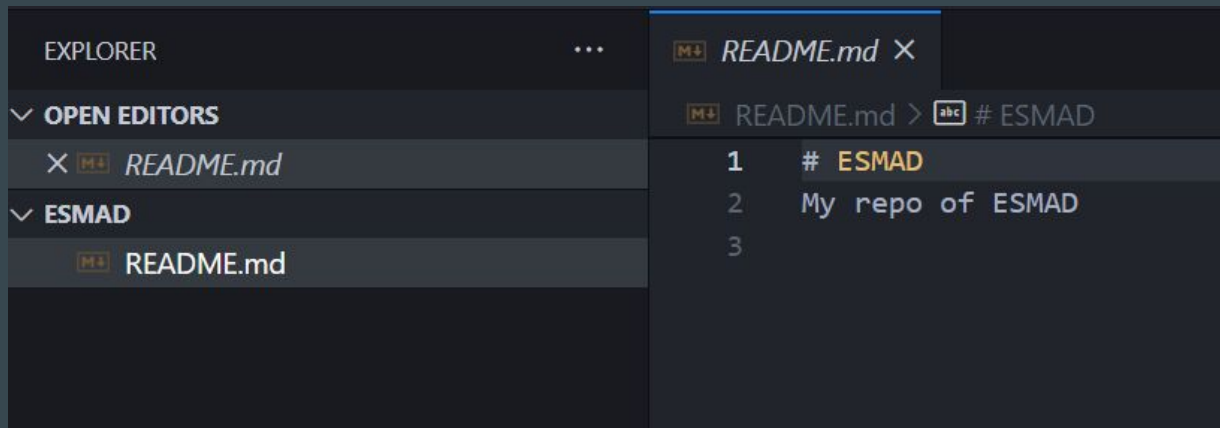
- Select the repository location at your computer
- Open the cloned repository



The Development Environment

2. Git & GitHub

6. Integrate the Repo in VSC
 - And the integration is made!

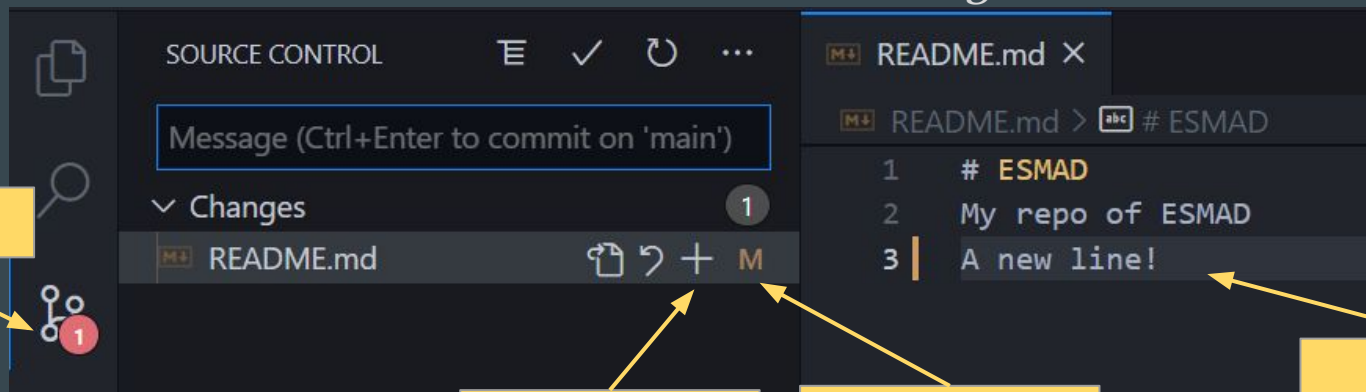


The Development Environment

2. Git & GitHub

6. Integrate the Repo in VSC

- Add a new line to the README file
- Go to the **Source Control** explorer
- Click in the “+” button to move the file to the stage area



State changes

Modified state

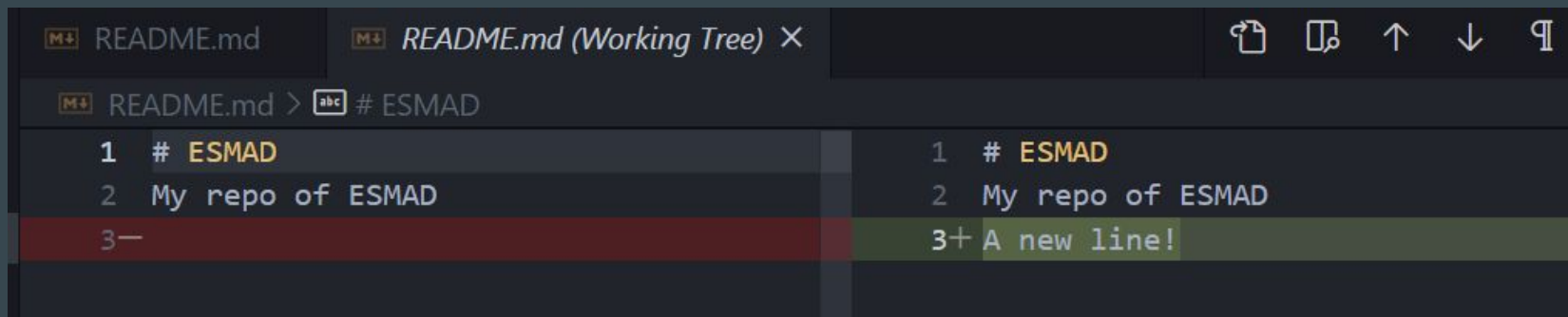
New line!

The Development Environment

2. Git & GitHub

6. Integrate the Repo in VSC

- If you click in the “M” icon you will see both files: the original and the modified file and with the differences highlighted!

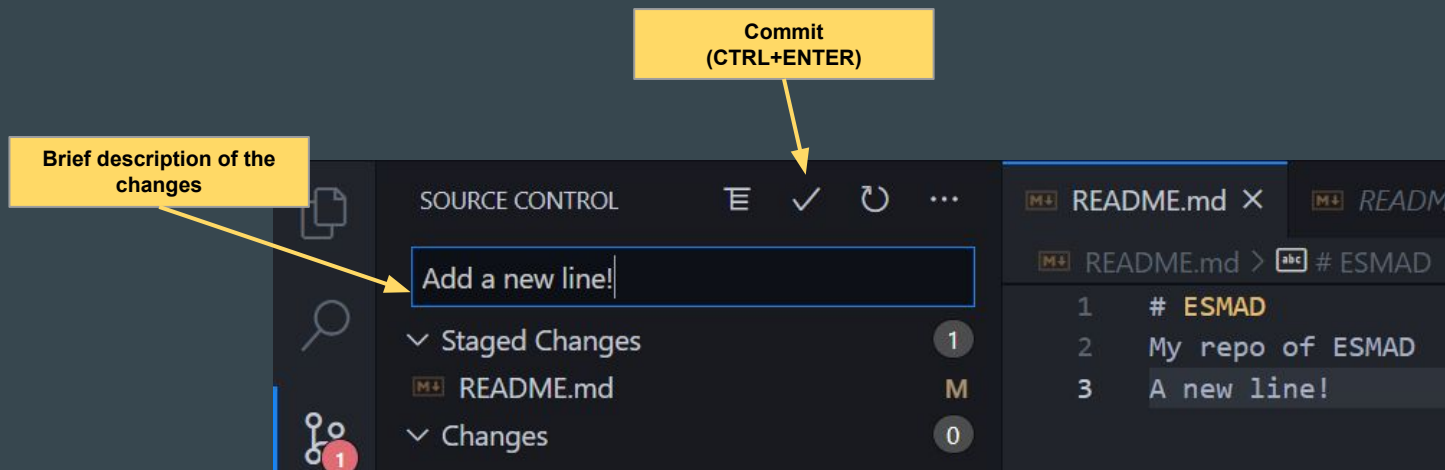


The Development Environment

2. Git & GitHub

6. Integrate the Repo in VSC

- Then, make commit to the Git local directory

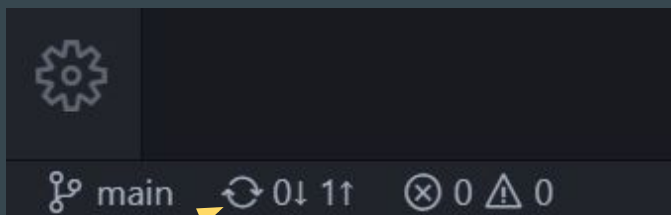


The Development Environment

2. Git & GitHub

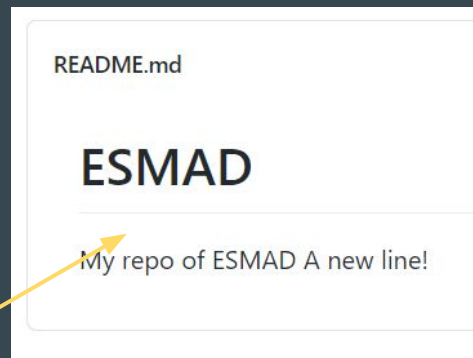
6. Integrate the Repo in VSC

- After commit, you must sync the changes with the remote server (GitHub) by pressing the button “**Synchronize changes**”
- To get new updates from the server you only need to press the same button



Synchronize button

File updated in the server!



The Development Environment

2. Git & GitHub

6. Integrate the Repo in VSC
 - Access to other options through the button “...”

